

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

“Information Science Engineering”

III, V & VII Semester

Dec/Jan - 2024

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	15MATDIP31	Additional Mathematics – I	Dec.23/Jan.24
2	BCS301	Mathematics for Computer Science	Dec.23/Jan.24
3	BCS302	Digital Design and Computer Organization	Dec.23/Jan.24
4	BCS303	Operating Systems	Dec.23/Jan.24
5	BCS304	Data Structures and Applications	Dec.23/Jan.24
6	21CS51	Automata Theory and Compiler Design	Dec.23/Jan.24
7	21CS52	Computer Networks	Dec.23/Jan.24
8	21CS53	Database Management Systems	Dec.23/Jan.24
9	21CS54	Artificial Intelligence and Machine Learning	Dec.23/Jan.24
10	18IS72	Software Architecture and Design Pattern	Dec.23/Jan.24
11	18CS71	Artificial Intelligence and Machine Learning	Dec.23/Jan.24
12	18CS72	Big Data and Analytics	Dec.23/Jan.24
13	18CS734	User Interface Design	Dec.23/Jan.24
14	18CS741	Digital Image Processing	Dec.23/Jan.24
15	18CS745	Robotic Process Automation Design and Development	Dec.23/Jan.24
16	18CS752	Python Application Programming	Dec.23/Jan.24

CBCS SCHEME

USN

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15MATDIP31

Third Semester B.E. Degree Examination, Dec.2023/Jan.2024 Additional Mathematics – I

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Show that

$$\frac{(\cos\theta + i \sin\theta)^4}{(\sin\theta + i \cos\theta)^4} = \cos 8\theta + i \sin 8\theta \quad (05 \text{ Marks})$$

- b. Prove that $[\vec{b} \times \vec{c}, \vec{c} \times \vec{a}, \vec{a} \times \vec{b}] = [\vec{a}, \vec{b}, \vec{c}]^2$ (05 Marks)

- c. Prove that $\hat{i} \times (\hat{a} \times \hat{i}) + \hat{j} \times (\hat{a} \times \hat{j}) + \hat{k} \times (\hat{a} \times \hat{k}) = 2\hat{a}$ (06 Marks)

OR

- 2 a. Find the real part of $\frac{1}{1 + \cos\theta + i \sin\theta}$ (05 Marks)

- b. If $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - 2\hat{k}$, find (i) $\vec{a} \cdot \vec{b}$ (ii) Angle between $\vec{a}$ & $\vec{b}$. (05 Marks)

- c. Show that $[\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}] = 2[\vec{a}, \vec{b}, \vec{c}]$ (06 Marks)

Module-2

- 3 a. If $y = \tan^{-1}x$, prove that $(1 + x^2)y_2 + 2xy_1 = 0$ and hence show that $(1 + x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$ (06 Marks)

- b. Find the angle between the radius vector and the tangent for the curve $r^m = a^m(\cos m\theta + \sin m\theta)$ (05 Marks)

- c. Find the pedal equation of $r^n = a^n \cos n\theta$ (05 Marks)

OR

- 4 a. If $u = \log\left(\frac{x^4 + y^4}{x + y}\right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$ (06 Marks)

- b. If $u = f(x - y, y - z, z - x)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ (05 Marks)

- c. Find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ where $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ (05 Marks)

Module-3

- 5 a. Evaluate the integral $\int_0^\pi x \sin^2 x \cos^4 x \, dx$ using Reduction formula. (05 Marks)

- b. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy \, dy \, dx$ (05 Marks)

- c. Evaluate $\int_{-1}^1 \int_0^2 \int_{x-2}^{x+2} (x + y + z) \, dy \, dx \, dz$ (06 Marks)

OR

- 6 a. Evaluate $\int_0^{\pi} \sin^4 x \, dx$, using Reduction formula. (05 Marks)
- b. Evaluate $\int_0^1 x^{3/2} (1-x)^{3/2} \, dx$, using Reduction formula. (06 Marks)
- c. Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) \, dy \, dx$, using Reduction formula. (05 Marks)

Module-4

- 7 a. A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$ where t is the time. Find the velocity and acceleration at any time t and also their magnitudes at $t = 0$. (05 Marks)
- b. Find $\text{div } \vec{F}$, where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ (06 Marks)
- c. Find the value of the constant 'a' such that the vector field $\vec{F} = (axy - z^3)\mathbf{i} + (a - 2)x^2\mathbf{j} + (1 - a)xz^2\mathbf{k}$ is irrotational. (05 Marks)

OR

- 8 a. If $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$ represents the parametric equation of the curve. Find the angle between the tangents at $t = 1$ and $t = 2$. (05 Marks)
- b. Find the angle between the normals to the surface at the points $(4, 1, 2)$ and $(3, 3, -3)$. (05 Marks)
- c. Show that $\vec{F} = \frac{x\mathbf{i} + y\mathbf{j}}{x^2 + y^2}$ is both solenoidal and irrotational. (06 Marks)

Module-5

- 9 a. Solve : $x^2y \, dx - (x^3 + y^3)dy = 0$ (06 Marks)
- b. Solve : $(2x + y + 1)dx + (x + 2y + 1)dy = 0$ (05 Marks)
- c. Show that $\frac{dy}{dx} - \frac{2y}{x} = x + x^2$ (05 Marks)

OR

- 10 a. Solve : $(y^3 - 3x^2y)dx - (x^3 - 3xy^2)dy = 0$ (05 Marks)
- b. Solve : $\frac{dy}{dx} + y \cot x = \cos x$ (05 Marks)
- c. Solve : $\frac{dy}{dx} + \frac{y}{x} = y^2x$ (06 Marks)

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Mathematics for Computer Science

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
 2. VTU Formula Hand Book is permitted.
 3. M : Marks, L: Bloom's level, C: Course outcomes.
 4. Mathematics hand book is permitted.

Module – 1									M	L	C																		
Q.1	a.	A Random variable X has the following probability function for variable values of x.							6	L2	CO1																		
		<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>P(x)</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k²</td><td>2k²</td><td>7k²+k</td></tr></table>										x	0	1	2	3	4	5	6	7	P(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k
		x	0	1	2	3	4	5				6	7																
P(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k																					
(i) Find the value of k. (ii) Evaluate P(x ≥ 6) and P(3 < x ≤ 6).																													
	b.	Find the mean and variance of Binomial distribution.							7	L2	CO2																		
	c.	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for, (i) 10 minutes or more. (ii) Less than 10 minutes. (iii) Between 10 and 12 minutes.							7	L3	CO2																		
OR																													
Q.2	a.	A random variable x has the following density function $P(x) = \begin{cases} Kx^2 & -3 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$. Find the value of K. Evaluate (i) P(1 ≤ x ≤ 2) (ii) P(x ≤ 2)							6	L2	CO1																		
	b.	In a factory producing blades, the probability of any blade being defective is 0.002. If blades are supplied in packets of 10, using Poisson distribution determine the number of packets containing, (i) No defective. (ii) One defective (iii) Two defective blades respectively in a consignment of 10,000 packets.							7	L2	CO2																		
	c.	In a test on electric bulbs, it was found that the life time of a particular brand was distributed normally with an average life of 2000 hours and standard deviation of 60 hours. If a firm purchases 2500 bulbs find the number of bulbs that are likely to last for, (i) More than 2100 hours. (ii) Between 1900 to 2100 hours. (iii) Less than 1950 hours. (Given φ(1.67) = 0.4525, φ(0.83) = 0.2967)							7	L3	CO2																		

Module – 2																													
Q.3	a.	The joint probability distribution table for two random variable x and y is as follows : <table><tr><td></td><td>Y</td><td>-2</td><td>-1</td><td>4</td><td>5</td></tr><tr><td>X</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td>0.1</td><td>0.2</td><td>0</td><td>0.3</td></tr><tr><td>2</td><td></td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr></table> Determine the marginal probability distribution of x and y. Obtain the correlation coefficient between x and y.		Y	-2	-1	4	5	X						1		0.1	0.2	0	0.3	2		0.2	0.1	0.1	0	6	L2	CO2
	Y	-2	-1	4	5																								
X																													
1		0.1	0.2	0	0.3																								
2		0.2	0.1	0.1	0																								
	b.	Find the unique fixed probability vector for the regular stochastic matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 6 & 2 & 3 \\ 0 & 2 & 1 \\ 3 & 3 & 3 \end{bmatrix}$	7	L2	CO3																								
	c.	Three boys A, B, C are throwing ball to each other. A always throws the ball to B and B always throws the ball to C. C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball find the probabilities that after three throws : (i) A has the ball. (ii) B has the ball. (iii) C has the ball.	7	L3	CO3																								
OR																													
Q.4	a.	The joint probability distribution of two discrete random variables x and y is given by $f(x, y) = k(2x+y)$ where x and y are integers. Such that $0 \leq x \leq 2, 0 \leq y \leq 3$. (i) Find the value of the constant K. (ii) Find the marginal probability distribution of X and Y. (iii) Show that the random variables X and Y are dependent.	6	L2	CO2																								
	b.	Find the unique fixed probability vector for the matrix, $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix}$.	7	L2	CO3																								
	c.	Each year a man trades his car for a new car in 3 brands of the popular company. If he has a 'swift' he trades it for 'Dzire'. If he has a 'Dzire' he trades it for a 'Wagnor'. If he has a 'Wagnor' he is just as likely to trade it for a new 'Wagnor' or for a 'Dzire' or a 'Swift' one. In 2020 he bought his first car which was 'Wagnor'. Find the probability that he has (i) 2022 Wagnor. (ii) 2022 Swift. (iii) 2023 Dzire. (iv) 2023 Wagnor.	7	L3	CO3																								
Module – 3																													
Q.5	a.	Explain the following terms: (i) Statistical Hypothesis. (ii) Critical region of statistical test. (iii) Test for significance.	6	L1	CO5																								

	b.	In 324 throws of a six faced die an odd number turned up 181 times. Is it reasonable to think that the die is an unbiased one at 5% level of significance?	7	L3	CO4																		
	c.	One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total 200 flights. Is there a significant difference in the two types of aircrafts so far as engine defects are concerned? Test at 5% significance level.	7	L3	CO4																		
OR																							
Q.6	a.	Define : (i) Null Hypothesis. (ii) Significance level. (iii) Type I and II error.	6	L1	CO5																		
	b.	A coin was tossed 1000 times and head turns up 540 times. Test the hypothesis that the coin is unbiased at 1% level of significance.	7	L3	CO4																		
	c.	In an exit poll enquiry it was revealed that 600 voters in one locality and 400 voters from an other locality favoured 55% and 48% respectively a particular party to come to power. Test the hypothesis that there is a difference in the locality in respect of the opinion at 1% level of significance.	7	L3	CO4																		
Module – 4																							
Q.7	a.	A random sample of size 64 is taken from an infinite population having mean 112 and variance 144. Using central limit theorem, find the probability of getting the sample mean $\bar{X}$ greater than 114.5	6	L2	CO5																		
	b.	The following data shows the runs scored by two batsman: Can it be said that the performance of batsman A is more consistent than the performance of batsman B? Use 1% level of significance ($F_{0.01, 4, 7} = 7.85$) <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>Batsman A</td><td>40</td><td>50</td><td>35</td><td>25</td><td>60</td><td>70</td><td>65</td><td>55</td></tr><tr><td>Batsman B</td><td>60</td><td>70</td><td>40</td><td>30</td><td>50</td><td>-</td><td>-</td><td>-</td></tr></table>	Batsman A	40	50	35	25	60	70	65	55	Batsman B	60	70	40	30	50	-	-	-	7	L2	CO4
Batsman A	40	50	35	25	60	70	65	55															
Batsman B	60	70	40	30	50	-	-	-															
	c.	A coins are tossed 100 times and the following results were obtained. Fit a binomial distribution for the data and calculate the theoretical frequencies. <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>Number of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Frequency</td><td>5</td><td>29</td><td>36</td><td>25</td><td>5</td></tr></table> (Given $\chi^2_{0.05} = 9.49$ for 4 degree of freedom)	Number of heads	0	1	2	3	4	Frequency	5	29	36	25	5	7	L3	CO4						
Number of heads	0	1	2	3	4																		
Frequency	5	29	36	25	5																		
OR																							
Q.8	a.	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find at 95% confidence interval for the population mean.	6	L2	CO4																		
	b.	The individuals are choosen at random from a population and their heights in inches are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71,71. Test the hypothesis that the mean height of the universe is 66 inches. (Given $t_{0.05} = 2.262$ for 9 degree of freedom).	7	L3	CO5																		
	c.	A sample analysis of examination results of 500 students war made. It was found that 220 students had failed, 170 had secured third class, 90 had secured second class and 20 had secured first class. Do these figures support the general examination result which is in the ratio 4 : 3 : 2 : 1 for the respective categories (Given $\chi^2_{0.05} = 7.81$ for 3 degree of freedom).	7	L3	CO4																		

Module – 5																																					
Q.9	a.	Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data : <table><tr><td>Food 1</td><td>8</td><td>12</td><td>19</td><td>8</td><td>6</td><td>11</td></tr><tr><td>Food 2</td><td>4</td><td>5</td><td>4</td><td>6</td><td>9</td><td>7</td></tr><tr><td>Food 3</td><td>11</td><td>8</td><td>7</td><td>13</td><td>7</td><td>9</td></tr></table>	Food 1	8	12	19	8	6	11	Food 2	4	5	4	6	9	7	Food 3	11	8	7	13	7	9	10	L3	CO6											
Food 1	8	12	19	8	6	11																															
Food 2	4	5	4	6	9	7																															
Food 3	11	8	7	13	7	9																															
	b.	Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz. A, B, C, D under a Latin-square design. <table><tr><td>C</td><td>B</td><td>A</td><td>D</td></tr><tr><td>25</td><td>23</td><td>20</td><td>20</td></tr><tr><td>A</td><td>D</td><td>C</td><td>B</td></tr><tr><td>19</td><td>19</td><td>21</td><td>18</td></tr><tr><td>B</td><td>A</td><td>D</td><td>C</td></tr><tr><td>19</td><td>14</td><td>17</td><td>20</td></tr><tr><td>D</td><td>C</td><td>B</td><td>A</td></tr><tr><td>17</td><td>20</td><td>21</td><td>15</td></tr></table>	C	B	A	D	25	23	20	20	A	D	C	B	19	19	21	18	B	A	D	C	19	14	17	20	D	C	B	A	17	20	21	15	10	L4	CO6
C	B	A	D																																		
25	23	20	20																																		
A	D	C	B																																		
19	19	21	18																																		
B	A	D	C																																		
19	14	17	20																																		
D	C	B	A																																		
17	20	21	15																																		
OR																																					
Q.10	a.	Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on four plots and state if the variety differences are significant at 5% significant level (Two way ANOVA). <table><tr><th rowspan="3">Plot of land</th><th colspan="3">Per acre production data</th></tr><tr><th colspan="3">Variety of wheat</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>1</td><td>6</td><td>5</td><td>5</td></tr><tr><td>2</td><td>7</td><td>5</td><td>4</td></tr><tr><td>3</td><td>3</td><td>3</td><td>3</td></tr><tr><td>4</td><td>8</td><td>7</td><td>4</td></tr></table>	Plot of land	Per acre production data			Variety of wheat			A	B	C	1	6	5	5	2	7	5	4	3	3	3	3	4	8	7	4	10	L3	CO6						
Plot of land	Per acre production data																																				
	Variety of wheat																																				
	A	B	C																																		
1	6	5	5																																		
2	7	5	4																																		
3	3	3	3																																		
4	8	7	4																																		
	b.	Set up ANOVA table for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people. <table><tr><th rowspan="2">Group of people</th><th colspan="3">Drug</th></tr><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td rowspan="2">A</td><td>14</td><td>10</td><td>11</td></tr><tr><td>15</td><td>9</td><td>11</td></tr><tr><td rowspan="2">B</td><td>12</td><td>7</td><td>10</td></tr><tr><td>11</td><td>8</td><td>11</td></tr><tr><td rowspan="2">C</td><td>10</td><td>11</td><td>8</td></tr><tr><td>11</td><td>11</td><td>7</td></tr></table> Do the drugs act differently? Are the different groups of people affected differently? Is the interaction term significant? Answer the above questions taking a significant level of 5%?	Group of people	Drug			X	Y	Z	A	14	10	11	15	9	11	B	12	7	10	11	8	11	C	10	11	8	11	11	7	10	L4	CO6				
Group of people	Drug																																				
	X	Y	Z																																		
A	14	10	11																																		
	15	9	11																																		
B	12	7	10																																		
	11	8	11																																		
C	10	11	8																																		
	11	11	7																																		

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BCS302

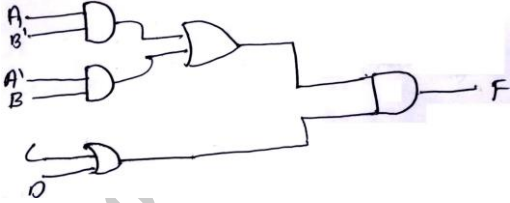
Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Obtain a minimum product of sums with a Karnaugh map. $F(w, x, y, z) = x' z' + wyz + w' y' z' + x' y.$		10	L3	CO1
	b.	Find the minimum sum of products for each function using a Karnaugh map i) $F_1(a, b, c) = M_0 + M_2 + M_5 + M_6$ ii) $F_2(d, e, f) = \sum m(0, 1, 2, 4)$ iii) $F_3(r, s, t) = rt' + r's' + r's$		10	L3	CO1
OR						
Q.2	a.	Identify the prime implicants and essential prime implicants of the following functions: i) $f(A, B, C, D) = \sum (1, 3, 4, 5, 10, 11, 12, 13, 14, 15)$ ii) $f(W, X, Y, Z) = \sum (0, 1, 2, 5, 7, 8, 10, 15).$		10	L3	CO1
	b.	Write the verilog code for the given expression using dataflow and behavioral model where $Y = (AB' + A'B) (CB + AD) (AB'C + AC).$		5	L2	CO1
	c.	Write the verilog code and time diagram for the given circuit with propagation delay where the AND, OR gate has a delay of 30ns and 10ns. 		5	L2	CO1
Module – 2						
Q.3	a.	What is Latch? With neat diagram, explain S-R latch using NOR gate. Derive characteristics equation.		10	L3	CO2
	b.	What is priority encoder? Design 4:2 priority encoder with necessary diagrams.		10	L3	CO2
OR						
Q.4	a.	Design and explain four bit adder with carry look ahead.		10	L3	CO2
	b.	What is multiplexer? Design 9:1 mux using 2:1 mux.		10	L3	CO2

BCS302					
Module – 3					
Q.5	a.	Explain four types of operation performed by computer with an example.	10	L2	CO3
	b.	Show how below expression will be executed in one address, two address zero address and three address processor in an accumulator organization $X = (A * B) + (C * D)$.	10	L1	CO3
OR					
Q.6	a.	What is addressing mode? Explain different types of addressing mode with an examples.	10	L2	CO3
	b.	With a neat diagram, explain basic operational concepts of a computer.	10	L2	CO3
Module – 4					
Q.7	a.	Explain the following with respect to interrupts with diagram. i) Vector interrupt ii) Interrupt nesting iii) Simultaneous request.	10	L2	CO3
	b.	Explain Direct Memory Access with a neat diagram.	10	L2	CO3
OR					
Q.8	a.	What is Bus arbitration? Explain different types of bus arbitration.	10	L2	CO3
	b.	Discuss different types of mapping functions of caches.	10	L2	CO3
Module – 5					
Q.9	a.	Draw and explain the single-bus organization of the data path inside a processor.	10	L2	CO4
	b.	List out the actions needed to execute the instruction ADD (R3), R1 write and explain the sequence of control steps for the execution of the same.	10	L2	CO4
OR					
Q.10	a.	Analyze how does execution of a complete instruction carry out.	10	L4	CO4
	b.	What is pipeline? Explain the performance of pipeline with an example.	10	L4	CO4

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Operating Systems

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M*: Marks, *L*: Bloom's level, *C*: Course outcomes.

Module – 1				M	L	C																				
Q.1	a.	Define Operating System. Explain dual mode of OS with a neat diagram.	5	L1, L2	CO1																					
	b.	Distinguish between the following terms: i) Multiprogramming and Multitasking ii) Multiprocessor system and clustered system.	10	L2	CO1																					
	c.	With a neat diagram, explain the concept the concept of VM-WARE architecture.	5	L1, L2	CO1																					
OR																										
Q.2	a.	Explain the operating system services with respect to programs and users.	5	L2	CO1																					
	b.	List and explain the different computing environments.	5	L1, L2	CO1																					
	c.	What are system calls? List and explain the different types of system calls.	10	L1, L2	CO1																					
Module – 2																										
Q.3	a.	Define process. Explain different states of a process with state diagram.	8	L1, L2	CO1																					
	b.	What is IPC? Explain direct and indirect communication with respect to message passing.	8	L1, L2	CO2																					
	c.	Explain context-switching.	4	L2	CO2																					
OR																										
Q.4	a.	What is multi-threaded process? Explain the four benefits of multithreaded programming.	6	L2	CO2																					
	b.	Calculate the average waiting time and average turn around time by drawing the Gantt-chart using FCFS, SJF-non preemptive, SRTF, RR(q = 2ms) and porosity algorithms.	14	L3	CO2																					
		<table> <tr> <th>Process</th> <th>Arrival time</th> <th>Burst time</th> <th>Porosity</th> </tr> <tr> <td>P1</td> <td>0</td> <td>9</td> <td>3</td> </tr> <tr> <td>P2</td> <td>1</td> <td>4</td> <td>2</td> </tr> <tr> <td>P3</td> <td>2</td> <td>9</td> <td>1</td> </tr> <tr> <td>P4</td> <td>3</td> <td>5</td> <td>4</td> </tr> </table>	Process	Arrival time	Burst time	Porosity	P1	0	9	3	P2	1	4	2	P3	2	9	1	P4	3	5	4				
Process	Arrival time	Burst time	Porosity																							
P1	0	9	3																							
P2	1	4	2																							
P3	2	9	1																							
P4	3	5	4																							
Module – 3																										
Q.5	a.	What is critical section? What are the requirements for the solution to critical section problem? Explain Peaterson’s solution.	8	L1, L2	CO3																					
	b.	Explain Reader’s-Writer’s problem using semaphores.	12	L2	CO3																					

1 of 2

OR

Q.6	a.	What is deadlock? What are the necessary conditions for the deadlock to occur?	6	L1, L2	CO3																																																																																										
	b.	Consider the following snap-shot of a system: <table><tr><th rowspan="2">Process</th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P0</td><td>2</td><td>0</td><td>0</td><td>1</td><td>4</td><td>2</td><td>1</td><td>2</td><td>3</td><td>3</td><td>2</td><td>1</td></tr><tr><td>P1</td><td>3</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>2</td><td>1</td><td>0</td><td>3</td><td>2</td><td>3</td><td>1</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td><td>4</td><td>2</td><td>4</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>1</td><td>4</td><td>3</td><td>2</td><td>3</td><td>6</td><td>6</td><td>5</td><td></td><td></td><td></td><td></td></tr></table> Answer the following using Banker's algorithm: i) Is the system in safe state? If so give the safe sequence. ii) If process P2 requests (0, 1, 1, 3) resource can it be granted immediately.	Process	Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P0	2	0	0	1	4	2	1	2	3	3	2	1	P1	3	1	2	1	5	2	5	2					P2	2	1	0	3	2	3	1	6					P3	1	3	1	2	1	4	2	4					P4	1	4	3	2	3	6	6	5					14	L3	CO3
Process	Allocation				Max				Available																																																																																						
	A	B	C	D	A	B	C	D	A	B	C	D																																																																																			
P0	2	0	0	1	4	2	1	2	3	3	2	1																																																																																			
P1	3	1	2	1	5	2	5	2																																																																																							
P2	2	1	0	3	2	3	1	6																																																																																							
P3	1	3	1	2	1	4	2	4																																																																																							
P4	1	4	3	2	3	6	6	5																																																																																							

Module – 4

Q.7	a.	What is paging? Explain with neat diagram paging hardware with TLB?	10	L1, L2	CO4
	b.	What are the commonly used strategies to select a free hole from the available holes?	6	L1	CO4
	c.	Explain fragmentation in detail.	4	L2	CO4

OR

Q.8	a.	With a neat diagram? Describe the steps in handling the page fault.	8	L2	CO4
	b.	Consider the page reference string: 1, 0, 7, 1, 0, 2, 1, 2, 3, 0, 3, 2, 4, 0, 3, 6, 2, 1 for a memory with 3 frames. Determine the number of page faults using F1, F0, optimal and LRU replacement algorithms which algorithm is more efficient.	12	L3	CO4

Module – 5

Q.9	a.	Define file. List and explain the different file attributes and operations.	10	L1	CO5
	b.	Explain the different allocation methods.	10	L2	CO5

OR

Q.10	a.	What is Access Matrix? Explain Access Matrix method of system protection with domain as objects and its implementation.	10	L1, L2	CO5
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request 143 and previously serviced a request at 125. The queue of pending requests in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by disk arm to satisfy the requests using FCFS, SSTF, SCAN, LOOK and C-LOOK algorithm.	10	L3	CO5

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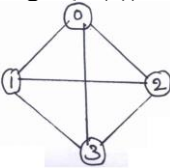
Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define Data Structures. Explain with neat block schematic different type of data structures with examples. What are the primitive operations that can be performed?	10	L2	CO1
	b.	Differentiate between structures and unions shown examples for both.	5	L1	CO1
	c.	What do you mean by pattern matching? Outline knuth, Morris, Pratt pattern matching algorithm.	5	L2	CO1
OR					
Q.2	a.	Define stack. Give the implementation of Push (), POP () and display () functions by considering its empty and full conditions.	7	L2	CO1
	b.	Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression 6, 2, /, 3, -, 4, 2, *, +	7	L3	CO1
	c.	Write the Postfix form of the following using stack : (i) $A*(B*C+D*E) + F$ (ii) $(a + (b*c) / (d-e))$	6	L3	CO1
Module – 2					
Q.3	a.	What are the disadvantages of ordinary queue? Discuss the implementation of circular queue.	8	L2	CO2
	b.	Write a note on multiple stacks and priority queue.	6	L2	CO2
	c.	Define Queue. Discuss how to represent queue using dynamic arrays.	6	L2	CO2
OR					
Q.4	a.	What is a linked list? Explain the different types of linked lists with neat diagram.	4	L2	CO2
	b.	Give the structure definition for singly linked list (SLL). Write a C function to, (i) Insert on element at the end of SLL. (ii) Delete a node at the beginning of SLL.	8	L3	CO2
	c.	Write a C-function to add two polynomials show the linked list representation of below two polynomials $p(x) = 3x^{14} + 2x^8 + 1$ $q(x) = 8x^{14} - 3x^{10} + 10x^6$	8	L3	CO2
Module – 3					
Q.5	a.	Write a C-function for the following operations on Doubly Linked List (DLL): (i) addition of a node. (ii) concatenation of two DLL.	8	L3	CO3
	b.	Write C functions for the following operations on circular linked list : (i) Inserting at the front of a list. (ii) Finding the length of a circular list.	8	L3	CO3

	c.	For the given sparse matrix, give the diagrammatic linked representation. $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 1 \\ 0 & 0 & 6 & 0 \end{bmatrix}$	4	L3	CO3
OR					
Q.6	a.	Discuss how binary tree are represented using, (i) Array (ii) Linked list	6	L2	CO3
	b.	Discuss inorder, preorder, postorder and level order traversal with suitable recursive function for each.	8	L2	CO3
	c.	Define Threaded Binary Tree. Discuss In-Threaded binary Tree.	6	L2	CO3
Module – 4					
Q.7	a.	Write a function to perform the following operations on Binary Search Tree (BST) : (i) Inserting an element into BST. (ii) Recursive search of a BST.	8	L3	CO4
	b.	Discuss selection Trees with an example.	8	L2	CO4
	c.	Explain Transforming a first into a binary tree with an example.	4	L2	CO4
OR					
Q.8	a.	Define graph. Show the adjacency matrix and adjacency list representation of the graph given below (Refer Fig. Q8 (a)).  Fig. Q8 (a)	6	L3	CO4
	b.	Define the following Terminologies with examples, (i) Digraph (ii) Weighted graph (iii) Self loop (iv) Parallel edges	8	L1	CO4
	c.	Explain in detail elementary graph operations.	6	L2	CO4
Module – 5					
Q.9	a.	What is collision? What are the methods to resolve collision? Explain linear probing with an example.	7	L2	CO5
	b.	Explain in detail, about static and dynamic hashing.	6	L2	CO5
	c.	Discuss Leftist Trees with an example.	7	L2	CO5
OR					
Q.10	a.	Explain different types of HASH function with example.	6	L2	CO5
	b.	Discuss AVL tree with an example. Write a function for insertion into an AVL Tree.	6	L3	CO5
	c.	Define Red-black Tree, Splay tree. Discuss the method to insert an element into Red-Black tree.	8	L2	CO5

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21CS51

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Automata Theory and Compiler Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms :
 i) String ii) Language iii) Alphabet iv) Length of string (04 Marks)
 b. Explain the various phases of compiler with neat diagram. (08 Marks)
 c. Define DFA and design a DFA to accept the following language:
 i) To accept strings having even number of a's and odd number of b's. (08 Marks)
 ii) To accept strings of a's and b's not having the substring aab. (08 Marks)

OR

- 2 a. Design the equivalent DFA to the following ϵ -NFA. (05 Marks)

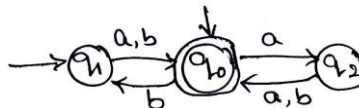
 b. Minimize the following DFA by identifying distinguishable and non-distinguishable states. (05 Marks)

δ	0	1
$\rightarrow$ A	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	H
H	G	C

- c. With neat diagram explain the components of language processing system in detail. (10 Marks)
(05 Marks)

Module-2

- 3 a. Define Regular Expressions. Write a regular expressions for the following :
 i) $L = \{a^n b^m \mid n+m \text{ is even}\}$
 ii) The set of all strings whose 3rd symbol from right end is 0
 iii) $L = \{a^{2n} b^{2m} \mid n \geq 0, m \geq 0\}$ (10 Marks)
 b. Convert the following automata to a regular expression. (04 Marks)



- c. Explain the concept of input buffering in the Lexical Analysis along with sentinels. (06 Marks)

OR

- 4 a. State and prove Pumping Lemma for regular languages and also prove the language $L = \{a^n b^n \mid n \geq 0\}$ is not a regular. (10 Marks)

- b. Construct ϵ -NFA for the following regular expression
 $(0 + 11) 0^* 1$ (04 Marks)
- c. Define Token, Lexeme and Pattern with example. (06 Marks)

Module-3

- 5 a. Define CFG. Write a CFG to the following languages.
 i) All strings over $\{a, b\}$ that are even and odd Palindromes.
 ii) $L = \{a^n \mid n \geq 0\}$ (10 Marks)
- b. Define ambiguity. Consider the grammar $E \rightarrow E + E \mid E * E \mid (E) \mid id$
 Construct the leftmost and rightmost derivation, parse tree for the string $id + id * id$.
 Also show that the grammar is ambiguous. (10 Marks)

OR

- 6 a. Consider the CFG given below with the production set, compute the following for the same.
 (i) First() and Follow() set (ii) Predictive Parsing table
 Grammar is,
 $E \rightarrow TE'$
 $E' \rightarrow +TE' \mid E$
 $T \rightarrow FT'$
 $T' \rightarrow *FT' \mid E$
 $F \rightarrow (E) \mid id$ (14 Marks)
- b. Write an algorithm to eliminate left recursion from a grammar. Also eliminate left recursion from the grammar
 $S \rightarrow Aa \mid b$
 $A \rightarrow Ac \mid Sd \mid \epsilon$ (06 Marks)

Module-4

- 7 a. Define PDA. Design PDA for the language $L = \{WCW^R \mid W \in (a, b)^*\}$ and also show the Instantaneous Description (ID) for the input $aabCb$. (10 Marks)
- b. Construct LR(0) automata for the grammar given below.
 $S \rightarrow L = R \mid R$
 $L \rightarrow *R \mid id$
 $R \rightarrow L$ (10 Marks)

OR

- 8 a. Define shift reduce Parser and Handle. Also list and explain the different actions operations available in Bottom up parser. (10 Marks)
- b. Construct the LR(1) automata for the given grammar.
 $S \rightarrow AA$
 $A \rightarrow aA \mid b$ (10 Marks)

Module-5

- 9 a. Design a Turing machine to accept the language $L = \{0^n 1^n 2^n \mid n \geq 1\}$ (10 Marks)
- b. Write a short note on the following :
 (i) Post correspondence problem (ii) Design issues in code generation (10 Marks)

OR

- 10 a. Translate the arithmetic expression $a = b * -c + b * -c$ into
 (i) Three address code (ii) Quadruple (iii) Triple (10 Marks)
- b. Write a short note on :
 (i) Decidable language (ii) Halting problems in Turing machines. (10 Marks)

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CBCS SCHEME

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21CS52

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Computer Networks. Explain local area network in detail with a neat diagram. (06 Marks)
- b. Explain MAN with a neat labelled diagram. (06 Marks)
- c. List and explain design issues for layer. (08 Marks)

OR

- 2 a. What are guided transmission media? Explain twisted pair cable in detail. (06 Marks)
- b. Explain TCP/IP reference model with a neat labelled diagram. (10 Marks)
- c. Briefly discuss virtual private networks. (04 Marks)

Module-2

- 3 a. List and explain any two data link layer design issues. (10 Marks)
- b. A bit stream transmitted using standard CRC method. The generator polynomial is $X^3 + 1$.
 - i) What is actual bit string transmitted
 - ii) Suppose 3rd bit from the left is inverted during transmission, how will receiver detect this error? (10 Marks)

OR

- 4 a. Explain Go-Back-N protocol working. (10 Marks)
- b. Briefly explain static channel and dynamic channel allocation problem. (10 Marks)

Module-3

- 5 a. Write an Dijkstra's algorithm to compute shortest path through graph. Explain with example. (10 Marks)
- b. Illustrate working of OSPF and BGP. (10 Marks)

OR

- 6 a. What is congestion control? List and explain various approaches to congestion control. (12 Marks)
- b. What is packet scheduling algorithm? Explain FIFO algorithm. (08 Marks)

Module-4

- 7 a. Write a program for congestion control using leaky bucket algorithm. (10 Marks)
- b. Briefly explain about transport service primitives. (10 Marks)

OR

- 8 a. With a neat labelled diagram, explain TCP segment structure. (10 Marks)
- b. Explain TCP connection management with TCP connection management FSM diagram. (10 Marks)

Module-5

- 9 a. Explain client/server and P-P architecture with a neat labelled diagram. (10 Marks)
b. Explain use and server interaction with a neat diagram. (10 Marks)

OR

- 10 a. Explain persistent and non persistent http in details. (10 Marks)
b. Write notes on:
(i) E-mail in the internet
(ii) Distributed DNS architecture (10 Marks)

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21CS53

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024

Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define DBMS. Explain all the basic operations that can be performed by DBMS on a database. (05 Marks)
- b. Explain the different users of a database system. (10 Marks)
- c. Describe the 3-Schema Architecture. (05 Marks)

OR

- 2 a. Define the following terms:
i) Data model ii) Schema iii) Insurance iv) Canned transaction (04 Marks)
- b. Describe the structural constraints of a database system with suitable example. (10 Marks)
- c. Explain all the E-R diagram notations. (06 Marks)

Module-2

- 3 a. Explain the four relational model constraints. (06 Marks)
- b. Explain all the steps of Relational database design using E-R to relational schema with a suitable example. (06 Marks)
- c. Discuss the DIVISION operation of relational algebra. Find the Quotient for the following :

A =

SNO	DNO
S ₁	P ₁
S ₁	P ₂
S ₁	P ₃
S ₁	P ₄
S ₂	P ₁
S ₂	P ₂
S ₃	P ₂
S ₄	P ₂
S ₄	P ₄

B₁ =

PNO
P ₂

B₂ =

PNO
P ₂
P ₄

B₃ =

PNO
P ₁
P ₂
P ₄

Find i) A/B₁ ii) A/B₂ iii) A/B₃ (08 Marks)

OR

- 4 a. Explain the characteristics of a relational model. (06 Marks)
- b. Explain all types of outer join operations in relational algebra. Demonstrate the advantage of outer join operation over the inner join operation. (06 Marks)
- c. Considering the following schema
Sailors (sid, sname, rating, age)
Boats (bid, bname, color)
Reserves (sid, bid, day)
Write a relational algebra queries for the following :
i) Find the names of sailors who have reserved boat#103.
ii) Find the names of sailors who have reserved a red boat.
iii) Find the names of sailors who have reserved a red or green boat.
iv) Find the names of sailors who have reserved all boats. (08 Marks)

Module-3

- 5 a. Explain the basic data types available for attributes in SQL. (05 Marks)
 b. Demonstrate the following constraints in SQL with suitable example:
 i) NOT NULL ii) Primary key iii) Foreign key iv) Default v) Check. (10 Marks)
 c. What are triggers? Explain with syntax and suitable example. (05 Marks)

OR

- 6 a. Explain the basic definition of a cursor and its usage with the help of a suitable example. (05 Marks)
 b. What are Assertions? Assuming suitable company schema write an Assertion for the condition.

“The salary of an Employee must not be greater than the salary of the manager of the department that the employee works for”. (05 Marks)

- c. Referring to the below mentioned company schema. Write the SQL queries for the following:
 Employee

Fname	Lname	Minit	Ssn	Bdate	Address	Sex	Salary	SuperSsn	Dno
-------	-------	-------	-----	-------	---------	-----	--------	----------	-----

Department

Dname	Dnumber	Mgr_Ssn	Mgr_start_date
-------	---------	---------	----------------

Department location

Dnumber	Dlocation
---------	-----------

Project

Pname	Pnumber	Plocation	Dnum
-------	---------	-----------	------

Work_on

Essn	DNo	HRS
------	-----	-----

Defendant

Essn	Dependentname	Sex	Bdate
------	---------------	-----	-------

- i) For each department retrieve the department number, the number of employees in the department and their average salary.
 ii) For each project on which more than 2 employees work, retrieve the project number, the project name and the number of employees who work on the project.
 iii) For each project, retrieve the project number, the project name and the number of employees from department no. 5 who work on that project.
 iv) For each department that has more than 5 employees, retrieve the department number and the number of its employees who are making more than \$40,000 salary.
 v) Retrieve the names of an employees who have two or more dependents. (10 Marks)

Module-4

- 7 a. Explain the types of update anomalies with examples. (05 Marks)
 b. Explain Armstrong's rules of inference. (05 Marks)
 c. What is the need for normalization? Explain 1NF, 2NF and 3NF with examples. (10 Marks)

OR

- 8 a. Explain the informal design guidelines of a database. (06 Marks)
 b. What is equivalence of sets of functional dependencies? Check whether the following sets of F.D's are equivalent or not.

$$FD_1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$$

$$FD_2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$$

- c. Write an algorithm to find the closure of functional dependency 'F'. (06 Marks)

Module-5

- 9 a. Explain the desirable properties of a transaction. (06 Marks)
 b. Explain with a neat diagram, the state transition diagram of a transaction. (06 Marks)
 c. Explain two phase locking mechanism with suitable example. (08 Marks)

OR

- 10 a. Discuss on the database inconsistency problem. (10 Marks)
 b. Explain Binary locks and shared locks with algorithms. (10 Marks)

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21CS54

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

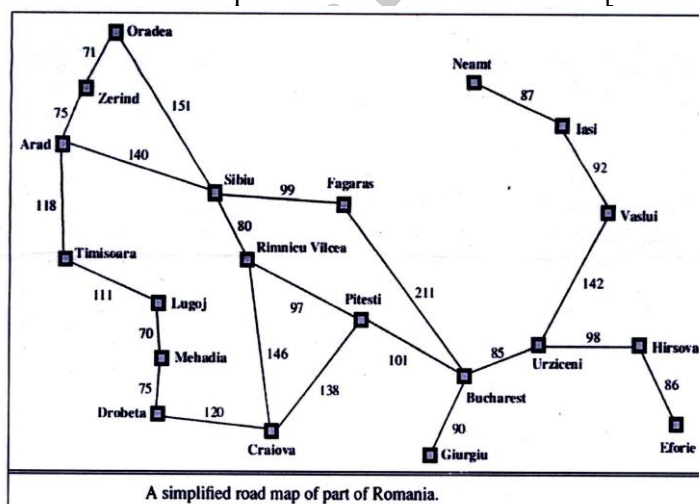
- 1 a. Identify the Turing test approach to provide a satisfactory operational definition of Intelligence. (04 Marks)
- b. Make use of the state space of the vacuum world and define the components to solve this problem. (06 Marks)
- c. Illustrate the properties and the algorithm for Breadth-first search technique. (10 Marks)

OR

- 2 a. Explain the concepts of thinking rationally and acting rationally. (04 Marks)
- b. Explain the tree search and graph search algorithms. (06 Marks)
- c. Explain problem solving agents alongwith the algorithm and illustrate the incremental formulation of 8-Queens problem. (10 Marks)

Module-2

- 3 a. Identify the differences between supervised and unsupervised learning. (04 Marks)
- b. Explain the types of Big data. (06 Marks)
- c. Apply A* algorithm to find the best path from Arad to Bucharest. [Refer Fig.Q3(c)].



Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

Values of h_{SLD} — straight-line distances to Bucharest.

Fig.Q3(c)

(10 Marks)

OR

- 4 a. Explain the machine learning process model along with diagram. (06 Marks)
- b. Consider the table given below which contains the machine learning course registration done by both boys and girls. There are 50 boys and 50 girls in the class and the registration of the course is given in the table. Apply Chi-square test and find out whether any differences exist between boys and girls for course registration.

Table 4(b)

Gender	Registered	Not Registered	Total
Boys	35	15	50
Girls	25	25	50
Total	60	40	100

(06 Marks)

- c. Apply the heuristic search algorithm on the given 8 puzzle problem to reach the goal state from the given initial state.

Initial State			Final State		
1	2	3	1	2	3
	4	6	4	5	6
7	5	8	7	8	

Fig.Q4(c)

(08 Marks)

Module-3

- 5 a. Consider the training dataset of 4 instances shown in the table below. Apply Find-S algorithm to find the final hypothesis.

Table 5(a)

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Logical Thinking	Interest	Job Offer
29	Yes	Excellent	Good	Fast	Yes	Yes
29	Yes	Good	Good	Fast	Yes	Yes
28	No	Good	Good	Fast	No	No
29	Yes	Good	Good	Slow	No	Yes

(08 Marks)

- b. Explain why Instance based learners are called lazy learners and compare instance based learning and model based learning. (06 Marks)
- c. Explain the types of Regression methods with diagram. (06 Marks)

OR

- 6 a. Consider the student performance training dataset of 8 data instances in the below table. Based on the performance of a student, classify the test instance (6.1, 40, 5) to check whether the student will pass or fail in that course using KNN approach ($K = 3$).

Table 6(a)

S.No.	CGPA	Assessment	Project Submitted	Result
1	9.2	85	8	Pass
2	8	80	7	Pass
3	8.5	81	8	Pass
4	6	45	5	Fail
5	6.5	50	4	Fail
6	8.2	72	7	Pass
7	5.8	38	5	Fail
8	8.9	91	9	Pass

(12 Marks)

- b. Explain version space and the candidate elimination algorithm explaining the algorithm steps. (08 Marks)

Module-4

- 7 a. Explain the advantages and disadvantages of decision trees. (06 Marks)
 b. Explain validating and pruning of decision trees. (06 Marks)
 c. Explain Bayes optimal classifier and solve to find whether a patient is diagnosed as COVID positive or COVID negative using the table given below.

Table 7(c)

$P(h_i/T)$	$P(\text{COVID positive})$	$P(\text{COVID negative}/h_i)$
0.3	0	1
0.1	1	0
0.2	1	0
0.1	1	0

(08 Marks)

OR

- 8 a. Explain the procedure to construct a decision tree using ID3 algorithm. (06 Marks)
 b. Explain Bayes theorem, Maximum A Posteriori (MAP) Hypothesis (h_{MAP}) and Maximum Likelihood (ML) Hypothesis (h_{ML}). (06 Marks)
 c. Illustrate the algorithm of Naïve Bayes and explain the popular variants of Bayesian classifier. (08 Marks)

Module-5

- 9 a. Explain the different activation functions used in ANN. (06 Marks)
 b. Illustrate the various types of Artificial Neural Networks. (08 Marks)
 c. Illustrate the applications and challenges of Clustering algorithms. (06 Marks)

OR

- 10 a. Explain the perceptron model and the algorithm. (08 Marks)
 b. Consider the following set of data given in the below table. Cluster it using K-means algorithm with the initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Table 10(b)

Objects	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

(12 Marks)

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15IS72

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Software Architecture and Design Pattern

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the General template of Design pattern. (08 Marks)
- b. Explain object oriented key concepts. (08 Marks)

OR

- 2 a. Explain the following :
i) Aggregation ii) Delegation iii) Inheritance iv) Abstract class. (08 Marks)
- b. Analyse various steps to use a design pattern. (08 Marks)

Module-2

- 3 a. Sketch the sequence diagram for adding a book into library system. (08 Marks)
- b. Explain guidelines to write Use Case. (08 Marks)

OR

- 4 a. Design class diagram for Boom class in a library system. (08 Marks)
- b. Write Use Case for place hold and remove hold. (08 Marks)

Module-3

- 5 a. Explain structure and participants of composite design pattern. (08 Marks)
- b. Describe Application and consequences of proxy design pattern. (08 Marks)

OR

- 6 a. Explain consequence and implementation of Decorator design pattern. (08 Marks)
- b. Write UML diagram to represent the structure and explain the participants of Flyweight Design pattern. (08 Marks)

Module-4

- 7 a. Discuss MVC architectural pattern. (08 Marks)
- b. Explain sequence diagram for drawing a line along with steps. (08 Marks)

OR

- 8 a. Discuss the scheme of implementing undo operation along with issues. (08 Marks)
- b. Explain the implementation details of Model class along with necessary prototype. (08 Marks)

Module-5

- 9 a. Explain Basic architecture of client/server systems. (04 Marks)
- b. List and briefly explain the steps to setting up a remote object. (12 Marks)

OR

- 10 a. Design state transition diagram for Add book and issue book. (07 Marks)
- b. Write a short note on:
i) Selection statement ii) Loop iii) Arrays (09 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

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18CS71

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Artificial Intelligence. What are the applications of Artificial Intelligence? (08 Marks)
- b. A water jug problem states "You are provided with two jugs, first one with 4-gallon capacity and the second one with 3-gallon capacity. Neither have any measuring markers on it. How can you get exactly 2-gallons of water into 4-gallon jug?"
 - i) Write down to production rules for the above problem.
 - ii) Write any one solution to the above problem. (12 Marks)

OR

- 2 a. Develop A* search algorithm for AI applications. (10 Marks)
- b. Explain problem characteristics with respect to heuristic search. (10 Marks)

Module-2

- 3 a. Explain the four approaches to knowledge representation. (10 Marks)
- b. Discuss the following set of sentences into WFF in predicate logic and hence find the answer with proof for the question whether the Marcus is loyal to Caesar or not.
 - i) Marcus was a man.
 - ii) Marcus was a Pompeian.
 - iii) All Pompeian's were Romans.
 - iv) Caesar was a ruler.
 - v) All Romans were either loyal to Caesar or hated him.
 - vi) Everyone is loyal to someone.
 - vii) People only try to assassinate rulers they are not loyal to.
 - viii) Marcus tried to assassinate Caesar.
 - ix) All mans are person. (10 Marks)

OR

- 4 a. Write Find S Algorithm and discuss issues with the algorithm. (10 Marks)
- b. Describe the Candidate Elimination algorithm. Find the maximum general hypothesis and maximum specific hypothesis for the training examples given in the table using candidate elimination algorithm.

Day	Sky	Air Temp.	Humidity	Wind	Water	Forecast	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Warm	Change	Yes

(10 Marks)

Module-3

- 5 a. Define decision tree. Construct the decision tree to represent the following Boolean functions:
 i) $A \wedge \neg B$ ii) $A \vee [B \wedge C]$ iii) $A \text{ XOR } B$ (06 Marks)
 b. Write the ID3 algorithm. (08 Marks)
 c. What do you mean by gain and entropy? How it is used to build the decision tree? (06 Marks)

OR

- 6 a. Define artificial neural networks. Discover what are the appropriate problems for neural network. (06 Marks)
 b. Define perceptron. Explain the concept of single perceptron with neat diagram. (06 Marks)
 c. Explain the back propagation algorithm with example. (08 Marks)

Module-4

- 7 a. What is Baye's theorem and maximum posterior hypothesis? (05 Marks)
 b. Derive an equation for MAP hypothesis using Baye's theorem. (05 Marks)
 c. Consider a football game between two rival teams: Team - A and Team - B. Suppose Team - A wins 95% of the time and Team - B wins the remaining matches. Among the games won by Team - A only 30% of them come from playing on Team - B's football field. On the other hand, 75% of the victories for Team - B are obtained while playing at home. If Team - B is to host the next match between the two teams which team will most likely emerge as the winner? (10 Marks)

OR

- 8 a. Discuss the Naïve Baye's classifier. (10 Marks)
 b. Discuss Minimum Description length principle in brief. (10 Marks)

Module-5

- 9 a. Explain K-nearest neighbor learning algorithm. (10 Marks)
 b. Define : i) Simple Error ii) True Error. (04 Marks)
 c. What is reinforcement learning? (06 Marks)

OR

- 10 a. Explain locally weighted linear regression. (10 Marks)
 b. Define expected value, variance standard deviation and estimate bias of a random variable. (05 Marks)
 c. Explain Q-learning with example. (05 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024
Big Data and Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. How is Data Architecture layers used for analytics? Explain with functions of each layer. (10 Marks)
- b. Briefly describe the three fundamental services offered by Cloud Computing. (10 Marks)

OR

- 2 a. List the features of Grid Computing. How does it differ from clusters and cloud computing. (10 Marks)
- b. Why is Data quality important in discovering new knowledge and decision making? (10 Marks)

Module-2

- 3 a. List Hadoop core components and explain with appropriate diagram. (10 Marks)
- b. Explain the working of the Hadoop Map Reduce frame work. (10 Marks)

OR

- 4 a. Explain the working of Hadoop – 2 Execution model (YARN Model). (10 Marks)
- b. With a diagram, explain the concept of APACHE Sqoop to acquire relational data. (10 Marks)

Module-3

- 5 a. Define NOSQL Explain Big Data NOSQL or Not – only SQL with its features, transactions and solutions. (10 Marks)
- b. Describe graph database characteristic, typical used and examples. (10 Marks)

OR

- 6 a. Explain Mongo DB with its features. (10 Marks)
- b. Compare and contrast RDBMS and Mongo DB databases. (05 Marks)
- c. What are the different ways of handling Big Data Problems? (05 Marks)

Module-4

- 7 a. Describe the Hive architecture components along with Hive Built – in functions. (10 Marks)
- b. Explain with respect to Hive QL :
 - i) Hive QL Data Definition Language (DDL).
 - ii) Hive QL Data Manipulation Language (DML). (10 Marks)

OR

- 8 a. Explain the architecture, feature and applications of PIG. (10 Marks)
- b. Illustrate by considering an example the working of the Map Reduce programming model. (10 Marks)

Module-5

- 9 a. How does regression analysis predict the value of the dependent variable in case of linear regression? (10 Marks)
b. Explain with example and algorithm, the working principle of Apriori process for adopting the subset of frequent item sets as a frequent itemset. (10 Marks)
- OR**
- 10 a. Define Web Mining. Discuss the broad classification of web mining and their applications. (10 Marks)
b. Define the term Social network. Explain social network as graphs with Centralities , Ranking and Anomaly Detection. (10 Marks)

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18CS734

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define GUI. Write the difference between GUI and webpage design. (10 Marks)
- b. Define user interface Design with example. Explain the importance and benefits of Good user Interface Design. (10 Marks)

OR

- 2 a. Discuss the general principles of UID. (10 Marks)
- b. Mention the advantages and disadvantages of GUI in details. (10 Marks)

Module-2

- 3 a. What is requirement analysis? What are the methods involved in it? What is the impact of it on UI design? (10 Marks)
- b. Define obstacles and pitfalls mention the general observation of design and common pitfalls and also explain five commandments used in Designing. (10 Marks)

OR

- 4 a. Explain the importance of human consideration in UI design with suitable example. (10 Marks)
- b. Explain briefly about human interaction speed. (10 Marks)

Module-3

- 5 a. Explain in brief the structure of Menu's. (10 Marks)
- b. Describe the components of a web navigation system with illustration. (10 Marks)

OR

- 6 a. Write a note on Graphical menus for the following
 - i) Pull down menu
 - ii) Pop up menu
 (10 Marks)
- b. Describe at least four guidelines to be followed in phasing of menu, during the development of system menus. (10 Marks)

Module-4

- 7 a. Discuss briefly about the types of windows with example. (Any five) (10 Marks)
- b. Write a note on the following
 - i) Track ball
 - ii) Joystick
 (10 Marks)

OR

- 8 a. Explain briefly about window management. (10 Marks)
- b. Write a note on components of a windows. (10 Marks)

Module-5

- 9 a. Explain briefly the following selection control
i) Radio buttons (10 Marks)
ii) Checkboxes (10 Marks)
- b. Explain the purpose of prototypes. Discuss any two kinds of prototypes with their importance to the system developers. (10 Marks)

OR

- 10 a. Explain the following with respect to kinds of Tests.
i) Think – Aloud Evaluation (10 Marks)
ii) Usability Test (10 Marks)
- b. Explain the types of presentation control. (10 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Digital Image with a neat block diagram, explain the fundamental steps of image processing. (08 Marks)
- b. Explain the applications of Image processing. (06 Marks)
- c. Let p, q be the two pixels with coordinates (100, 120) and (130, 160) respectively. Compute:
 - i) Euclidean distance
 - ii) Chess board distance
 - iii) Manhattan distance. (06 Marks)

OR

- 2 a. Explain the data structure for representing digital image. (08 Marks)
- b. Differentiate between sampling and quantization. (06 Marks)
- c. Find the shortest 4, 8 m-path between p and q consider $v = \{0, 1\}$

	3	1	2	1	q
	2	2	0	2	
	1	2	1	1	
p	1	0	1	1	

(06 Marks)

Module-2

- 3 a. Explain basic gray level transformation techniques. (08 Marks)
- b. Explain image smoothing spatial filter and sharpening spatial filter. (06 Marks)
- c. Apply logarithmic transformation on the input image by considering $C = 1$.

110	120	90
91	94	98
90	91	99

(06 Marks)

OR

- 4 a. Explain image enhancement using arithmetic and logic operations. (08 Marks)
- b. Explain image enhancement using first and second order derivatives. (06 Marks)
- c. Consider a gray scale image in matrix form. Let each element be a pixel of an image and values of the element represent intensities of the pixel. Perform Histogram equalization on the image and scale the intensity from 1 to 20.

3	2	4	5
7	7	8	2
3	1	2	3
5	4	6	7

(06 Marks)

Module-3

- 5 a. What is Sampling theorem? Explain sampling and Fourier transform of sampled function. (10 Marks)
 b. State and explain properties of 2D Fourier transform function. (10 Marks)

OR

- 6 a. Explain the basic steps for filtering in the frequency domain. (10 Marks)
 b. Explain Sharpening in the frequency domain using high pass filter. (10 Marks)

Module-4

- 7 a. What are the three basic objectives of canny edge detector? Explain canny edge detection algorithm. (08 Marks)
 b. Explain point, edge and line detection techniques. (06 Marks)
 c. Explain Mar-Hildreth edge detection algorithm in detail. (06 Marks)

OR

- 8 a. Optimal global thresholding can be achieved by Otsu's method. Justify the answer with an algorithm. (08 Marks)
 b. Generate the Sobel and Prewitt masks for the matrix given below :

$$G_x = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

- c. What is meant by edge linking? Explain edge linking using local processing. (06 Marks)

Module-5

- 9 a. Define Image Compression. Explain coding spatial and temporal redundancy. (06 Marks)
 b. Using arithmetic coding, decode the message 0.572 given the coding model

Symbol	!	C	E
Probability	0.1	0.4	0.5

- c. Explain Image, compression system with a neat diagram. (08 Marks)

OR

- 10 a. Explain Lempel-Ziv-Welch (LZW) coding with an example. (06 Marks)
 b. The character a to h have the set of frequencies based on the first 8 Fibonacci numbers as follows :
 a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21
 A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code (08 Marks)
 110111100111010
 c. Explain Run length coding with an example. (06 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Robotic Process Automation Design and Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List and explain the benefits of RPA. (10 Marks)
- b. Discuss the different flavours of RPA in detail. (05 Marks)
- c. Explain OCR skill in RPA. (05 Marks)

OR

- 2 a. Discuss about the downsides of RPA. (10 Marks)
- b. Discuss the on-premise Vs cloud RPA skill. (05 Marks)
- c. Explain the flowchart in RPA. (05 Marks)

Module-2

- 3 a. Explain the UiPath RPA component with diagram. (10 Marks)
- b. Discuss the emptying of recycle bin process with process flow diagram with activities. (10 Marks)

OR

- 4 a. List and explain different RPA platforms. (10 Marks)
- b. Discuss different types of task recording methods in UiPath. (10 Marks)

Module-3

- 5 a. Explain different types of project available in UiPath. (10 Marks)
- b. Discuss about building database dynamically. (10 Marks)

OR

- 6 a. List and explain different types of control flow activities available in UiPath. (10 Marks)
- b. Explain different activities used for Excel file activity. (10 Marks)

Module-4

- 7 a. Explain any five activities that help in finding the controls. (10 Marks)
- b. Discuss the screen scraping in UiPath. (10 Marks)

OR

- 8 a. Explain different types of events in UiPath. (10 Marks)
- b. Discuss three techniques for waiting for a control. (05 Marks)
- c. Explain keyboard activities in UiPath. (05 Marks)

Module-5

- 9 a. Discuss about exception Handling in UiPath. (10 Marks)
- b. Explain error reporting activity. (05 Marks)
- c. Discuss about collecting crash dump in UiPath. (05 Marks)

OR

- 10 a. Explain any three common exceptions with ways to handle them in UiPath. (09 Marks)
- b. Discuss about Chatbots. (05 Marks)
- c. Explain client logging and server logging in UiPath. (06 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

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18CS752

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Python Application Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Distinguish between :
 - i) Interpreter and compiler
 - ii) Syntax, logic and Semantic errors. (08 Marks)
- b. Define variable and keyword. List the keywords and rules followed to use variable names with an example. (06 Marks)
- c. Write a note on :
 - i) Accepting input from keyboard
 - ii) Choosing mnemonic variable names. (06 Marks)

OR

- 2 a. Demonstrate the conditional, alternative, chained and nested execution statements with syntax, flowchart and example. (12 Marks)
- b. Develop a student grading program that takes score with value between 0.0 to, 1.0 as its parameter and returns a grade as string. If score is out of range/string, then print appropriate error message using try and except concept.

Score	Grade
≥ 0.9	A
≥ 0.8	B
≥ 0.7	C
≥ 0.6	D
< 0.6	F

(08 Marks)

Module-2

- 3 a. Explain definite and indefinite loops with suitable examples. (05 Marks)
- b. Develop a program which repeatedly reads numbers until the user enters "done". Once "done" is entered, print total, count, average, maximum and minimum of numbers. Use try and except to print appropriate error message and skip to next number input. (10 Marks)
- c. List out the string handling methods with syntax and examples. (05 Marks)

OR

- 4 a. Describe the concept of parsing Strings and Format operators with suitable examples. (04 Marks)
- b. Demonstrate the open, read, write, search and close file methods with syntax and examples. (12 Marks)
- c. Develop a program to create a string made of first, middle and last character of a user specified string. (Ex : Input, : James, Output : Jms). (04 Marks)

Module-3

- 5 a. Illustrate 8 list handling methods in python. (08 Marks)
b. Bring out the relationship between list and functions with suitable examples. (06 Marks)
c. Development a program to turn every item of a list 1 into its square and place it into list 2. Print both lists. (06 Marks)

OR

- 6 a. Develop a program to read through a word file, find out the frequency of words in a file by ignoring the punctuation and alphabet case using dictionary. Print error if file does not exist. (08 Marks)
b. Describe tuple assignment with examples. (06 Marks)
c. Illustrate searching and extracting operational methods using regular expression. (06 Marks)

Module-4

- 7 a. Define instantiation. Explain the shallow and deep copy concept with examples. (08 Marks)
b. Demonstrate the concept of sameness between instances and use of instances as arguments and return values. (12 Marks)

OR

- 8 a. Illustrate the concept of pure functions and modifiers. (05 Marks)
b. Develop a program with initialization method and optional arguments. (10 Marks)
c. Demonstrate operator overloading and polymorphism feature with sample code. (05 Marks)

Module-5

- 9 a. Develop a program that represent World's simplest web browser. Also draw a conceptual diagram. (07 Marks)
b. Develop a program that can read any size file without using up all the memory in computer. (06 Marks)
c. Demonstrate the XML and JSON formats for data exchange across the web. (07 Marks)

OR

- 10 a. Write a note on Google geo-coding API web service. (08 Marks)
b. Demonstrate the use of CREATE, INSERT, SELECT, UPDATE and DELETE SQL commands in python. (12 Marks)

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